

Brain-wide neural activity during a decision-making task

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Allen Institute / Neural Dynamics

University of Puget Sound NRSC 490

09/23/26



alleninstitute/

Outline

1. Overview of Allen Institute / Neural Dynamics accelerator
2. Dynamic Routing dataset
 1. Background and motivation
 2. Experimental Methods
 3. Data overview
 4. Data Portal Demo
3. Hands-on code tutorial (Google Colab)
4. Wrap up

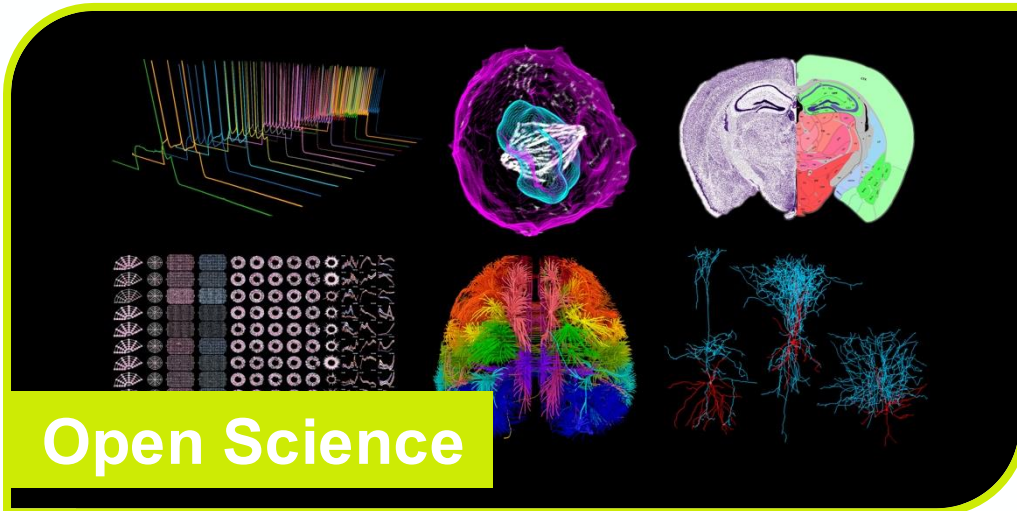
Allen Institute

- **Founding date:** September 16, 2003
- **Founders:** Paul G. Allen; Jody Allen
- **Employees:** ~900
- **Non-Profit**
- **Foundational Science**

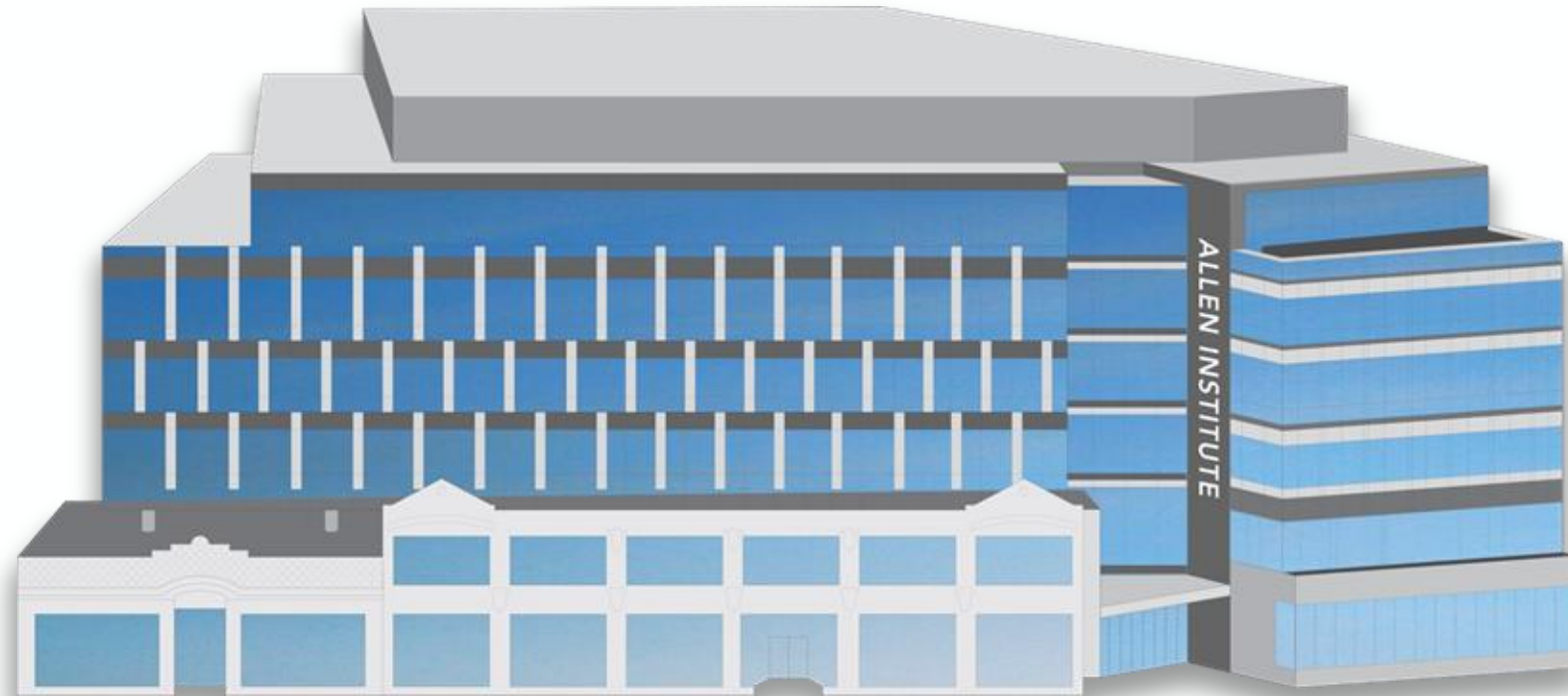
understandinglife
advancinghealth



Core Principles



Allen Institute, many accelerators



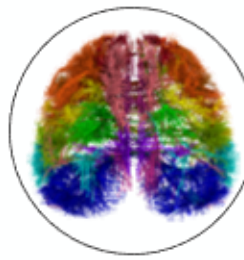
Allen Institute
for Brain
Science



Human Brain
Atlas



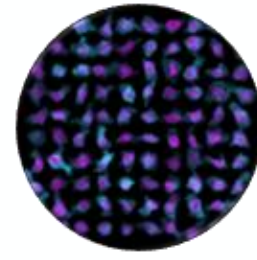
Connectivity
Atlas



The Paul G.
Allen Frontiers
Group



Allen Cell
Explorer



Alzheimer's Brain
Cell Atlas



Human Immune
System Explorer



Seattle Hub for
Synthetic Biology



2003



2025



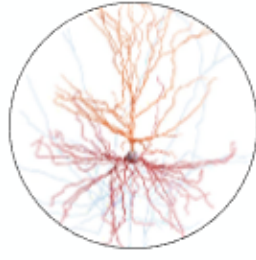
Mouse Brain
Atlas



MindScope



Allen Institute
for Cell
Science



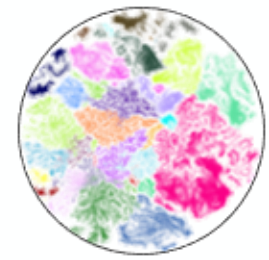
Cell Types Database
&
Brain Observatory



Allen Institute
for
Immunology



Allen Institute
for Neural
Dynamics



Allen Brain
Cell Atlas

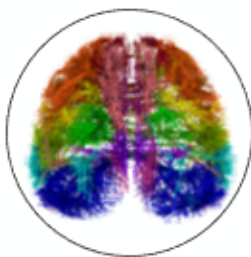
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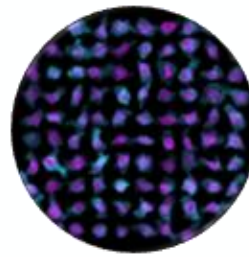
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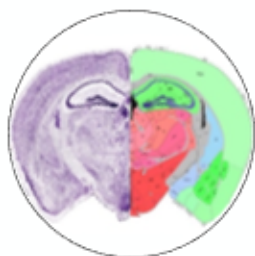
Seattle Hub for
Synthetic Biology



2003



2025



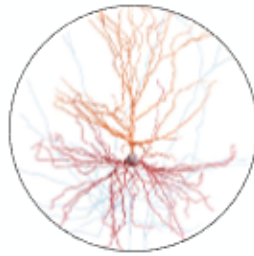
Mouse Brain
Atlas



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Allen Institute
for Cell
Science



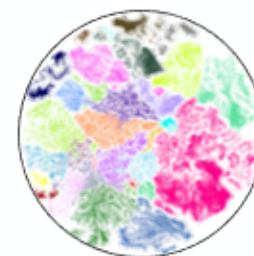
Cell Types Database
&
Brain Observatory



Allen Institute
for
Immunology



Allen Institute
for Neural
Dynamics



Allen Brain
Cell Atlas

We explore the brain's circuits and activity

Discovering how information processed by networks of neurons
generates our thoughts, emotions, and actions

**Multi-color imaging of centimeter-scale
specimens at nanoscale resolution.**

This ExA-SPIM image is one of the 15 tiles
used to acquire the entire volume of the
mouse brain. The field of view is focused on
the cerebellum.

Discovering how information processed by
networks of neurons generates our
thoughts, emotions, and actions

From individual neurons to the whole brain, we work to reveal how the brain interprets environments and make decisions



Allen Institute / Neural Dynamics Overview

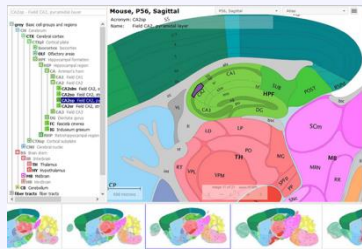
Leveraging *ex vivo* brain map, connectivity, and cell type atlases

Catalog of cell types based on anatomy, connectivity, and genes. Measured from extracted cells *ex vivo*

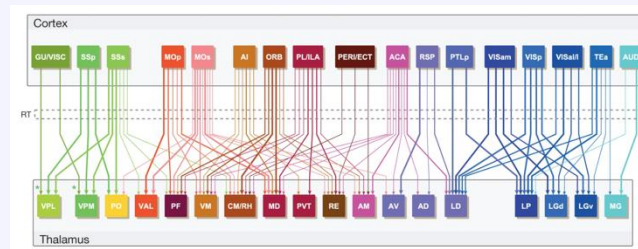
Allen Institute / Neural Dynamics Overview

Leveraging ex vivo brain map, connectivity, and cell type atlases

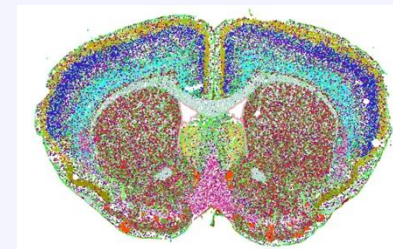
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Ng et al 2009, *Nature Neuroscience*



Oh, Harris, Ng, et al 2014, *Nature*

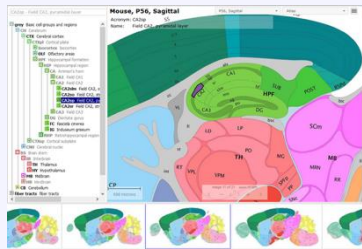


Yao et al., 2023, *Nature* (cells color-coded by cell type)

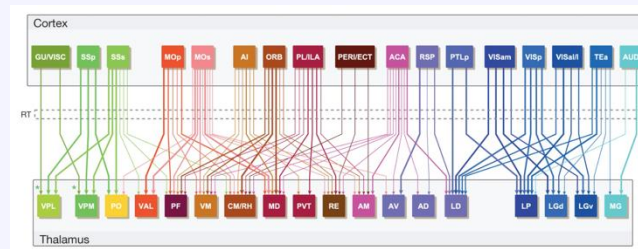
Allen Institute / Neural Dynamics Overview

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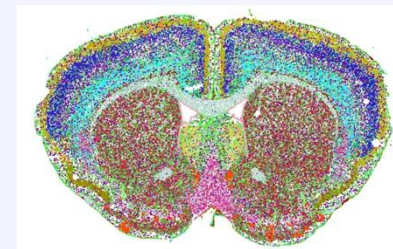
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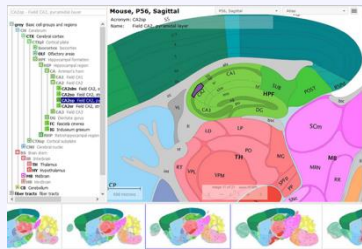


Understanding how those cells function *in vivo*

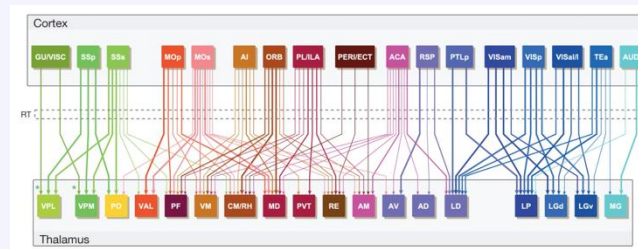
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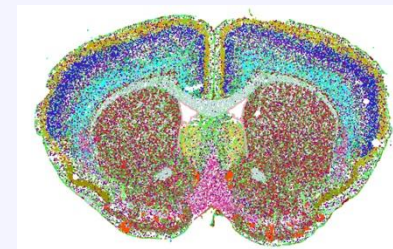
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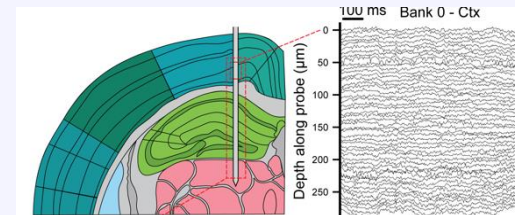


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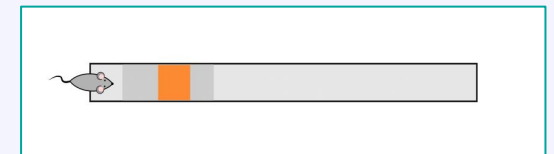


Understanding how those cells function *in vivo*

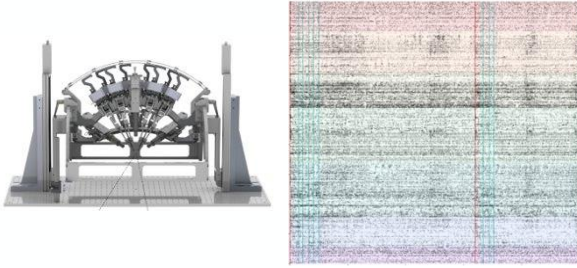
- Record from the same cell types in living, behaving animals
- Whole-brain scale: population activity at single-cell resolution
- Complex behavior: decision making, learning, foraging



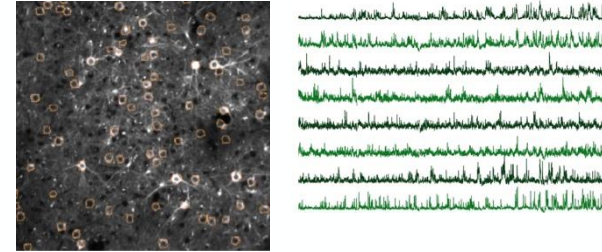
Ye & Shelton et al., 2025, *Neuron*



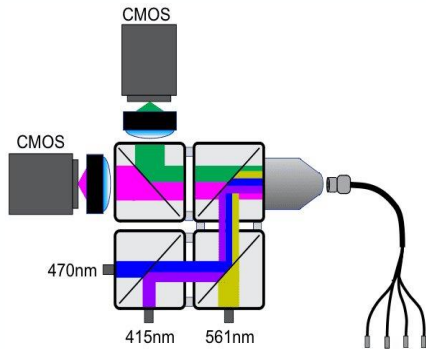
Standardized data acquisition platforms



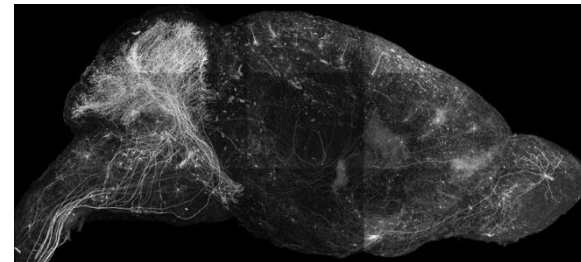
Electrophysiology



Cellular imaging



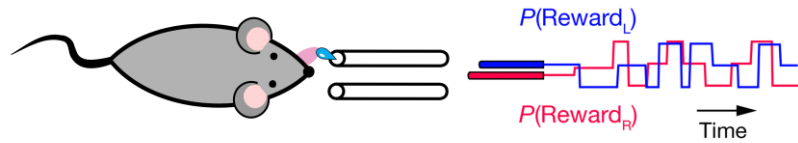
Fiber photometry



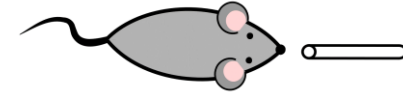
Lightsheet Imaging

Modular Behavior Platform

Dynamic Foraging



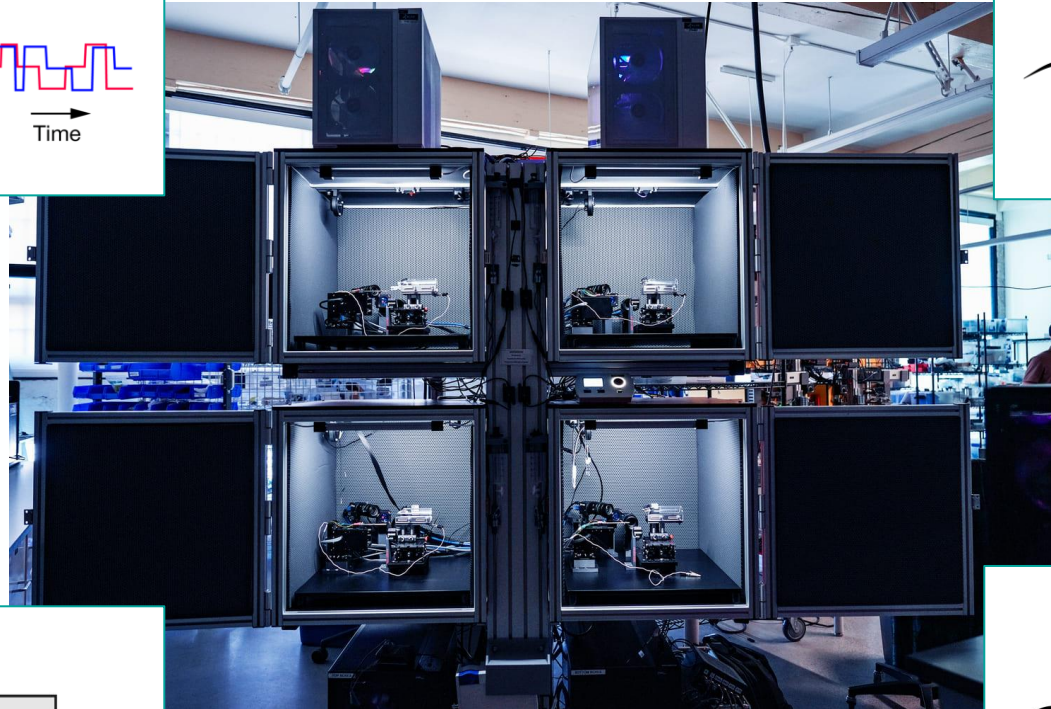
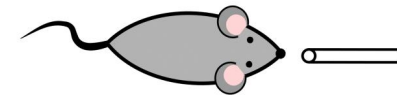
Change Detection



Patch Foraging (VR)



Dynamic Routing





Data Processing and Open Data

Neural Dynamics generates hundreds of terabytes of data per week!

All of this data is open and publicly accessible!

Data Assets

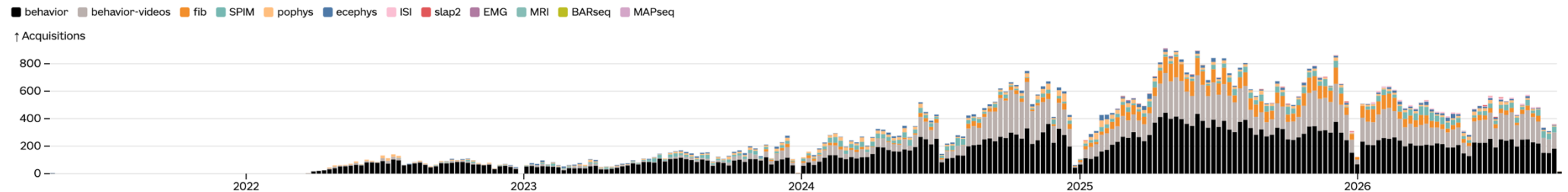
▼ DATA OVERVIEW

112,111 Total assets

6,127 Subjects

91 Projects

97 Investigators





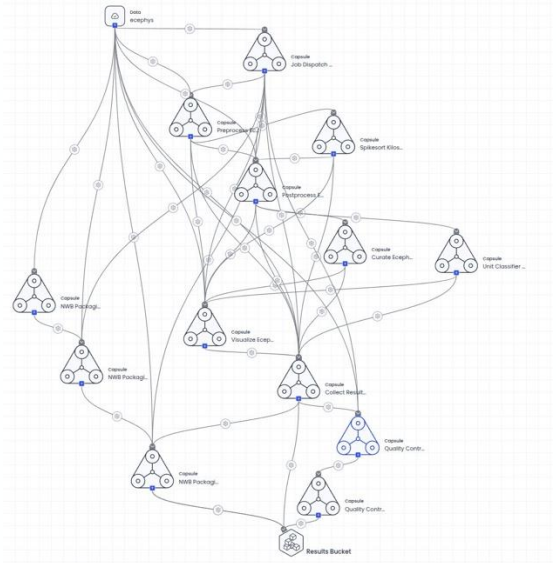
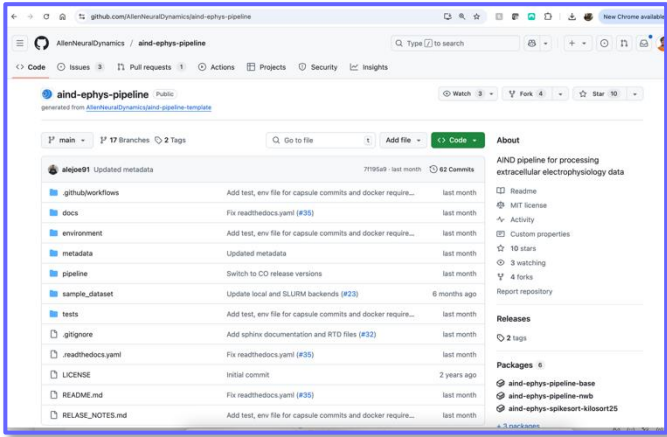
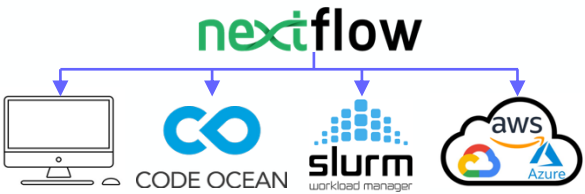
Scalable data processing pipelines

Accessible

Standardized

Transparent

Reproducible



 **aind-ephy-pipeline** Public 

 **aind-multiplane-ophys-pipeline** Public 

 **aind-smartspim-pipeline** Public 

 **behavior-video-inference-pipeline** Public 

 **aind-fiber-photometry-pipeline** Public 

A data platform for collaboration



Accessible

Standardized

Transparent

Reproducible



Open Data
on AWS Data
Exchange

Public at upload
Raw and processed



GitHub

Code in public repos



protocols.io



Metadata
Catalog

JSON Document DB
Easy to search



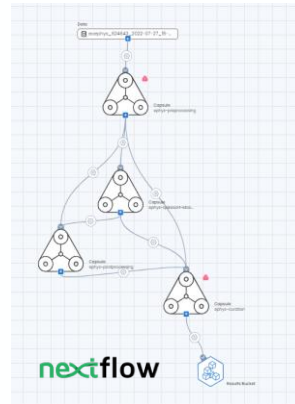
Data Portals

Data visualizations
QC evaluations
Reporting and browsing

Reproducible Run



cloud analysis



CODE OCEAN

Team Science Projects

Behavior and Motor Control

We are developing and using genetic, electrophysiological, optical, and behavioral approaches to investigate how the brain adaptively controls behavior. The team focuses on understanding the descending circuits that control the execution of actions and how they change when actions are reinforced and refined.

[→ Explore Project](#)

Brain-Wide Circuit Dynamics

We are using large-scale electrophysiology to study how distributed brain regions coordinate their spiking activity to guide behavior in changing environments.

[→ Explore Project](#)

Neural Dynamics in Multi-Regional Circuits with Thalamus in the Middle

We are uncovering how neural signals are transmitted from subcortical brain regions via the thalamus to modulate cortical activity and thereby influence behavior.

[→ Explore Project](#)

Cell Types & Learning

We are combining longitudinal in vivo calcium imaging with spatial transcriptomics to investigate how specific cell types restructure their activity during novelty processing and task learning.

[→ Explore Project](#)

Dynamic Routing

We are studying task-switching behaviors in mice to determine how the brain controls the flow of its own activity and how neuronal circuits are reconfigured to dynamically route information for different tasks.

[→ Explore Project](#)

OpenScope

OpenScope provides access to cutting-edge neurophysiological methods to scientists across the world. Similar to astronomical observatories, scientists propose experiments that are then executed at the Allen Institute.

[→ Explore Project](#)

Single-Cell Computation

Our goal is to investigate the computations performed by individual neurons, based on measuring synaptic input and

Brain-Wide Neuromodulation

We are studying the molecular and anatomical subclasses of neuromodulator neurons and exploring how networks use

Credit Assignment During Learning

We are using optical connection-mapping techniques and brain computer interface (BCI) in the context of a learning task to ask

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Dynamic Routing Project

Dynamic Routing Project

Project Team



Shawn Olsen, Investigator



Corbett Bennett, Scientist III



Shailaja Akella, Scientist II



Ethan McBride, Scientist II



Sam Gale, Scientist III



Ben Hardcastle, Scientist II



Vayle LaFehr, Research Associate II



Hannah Cabasco, Research Associate II

**how is information routed between
brain regions in a context-dependent
manner?**

context switching

Dynamic Routing Project

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**how is information routed between
brain regions in a context-dependent
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context switching

context-dependent behavior



context-dependent behavior



Context 1: Human present



Behavior: 'good dog'

Context 2: Human absent

context-dependent behavior



Context 1: Human present



Behavior: 'good dog'

Context 2: Human absent

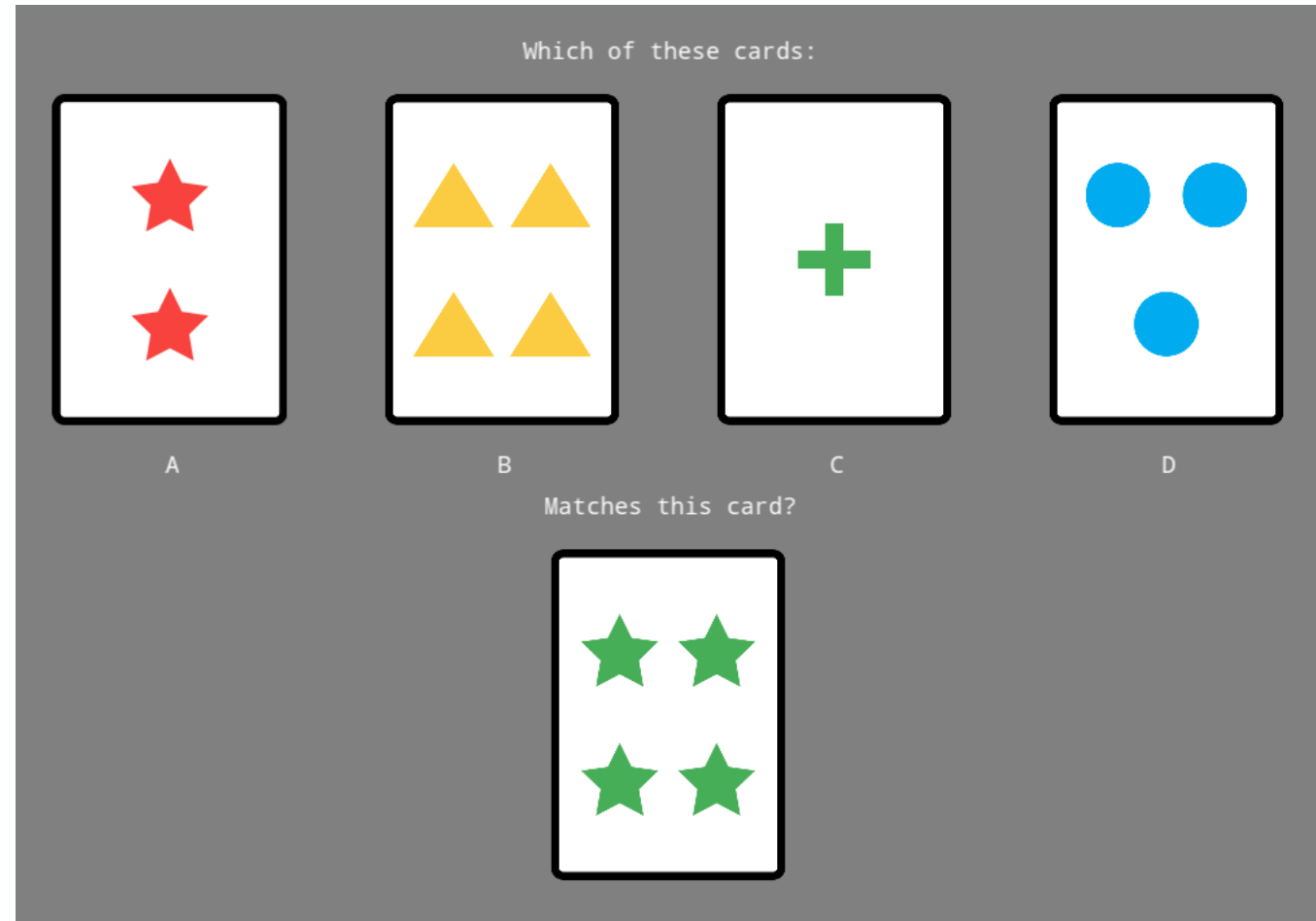


Behavior: 'bad dog'

Dynamic Routing Behavior task



Wisconsin Card Sorting Test



Wisconsin Card Sorting Test

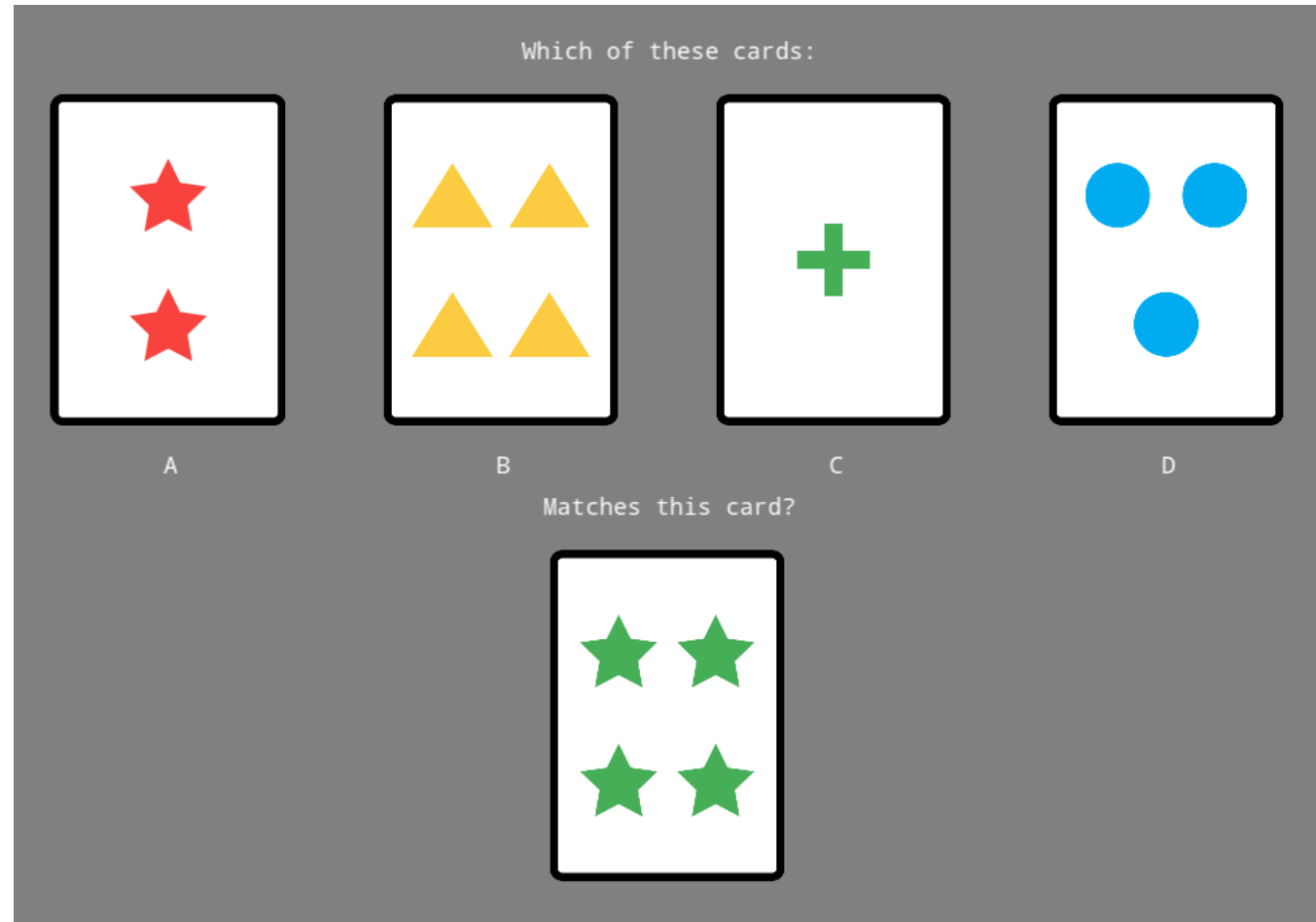
Task/rule shifting

Tests:

Cognitive flexibility

Abstract reasoning

Working memory



Effects of Different Brain Lesions on Card Sorting

The Role of the Frontal Lobes

MILNER, B., 1963. Archives of Neurology 9, 90–100.

BRENDA MILNER, PhD
MONTREAL

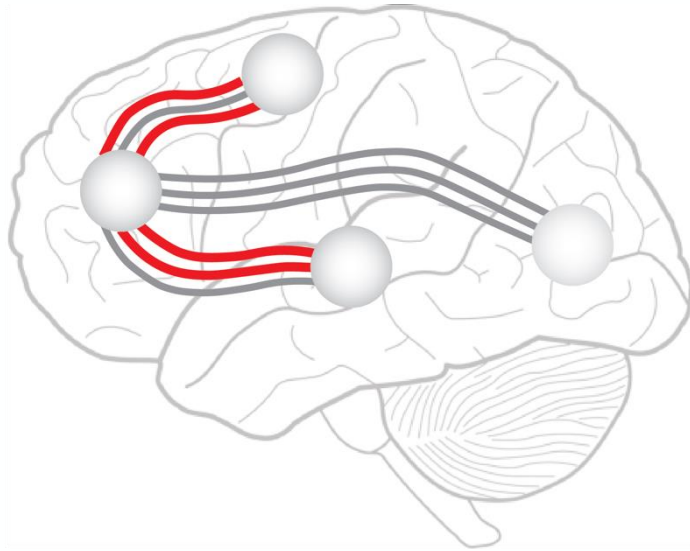


frontal cortex removal impairs task switching

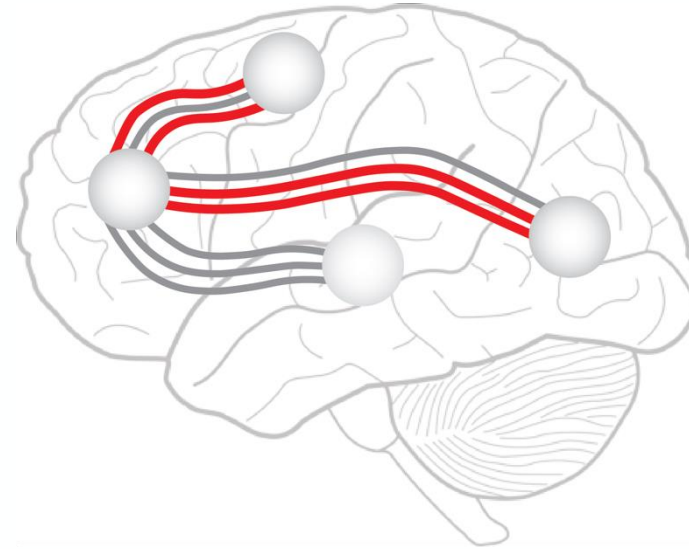
		Categories Achieved, Max 6			
			Preop.	Postop.	
		No.			
		Cases	Mean		
	Locus of Excision				
F8		Dorsolateral frontal	18	3.3	1.4
T18		Total control	53	4.6	4.7

Dynamic Routing Behavior task

listening (auditory)

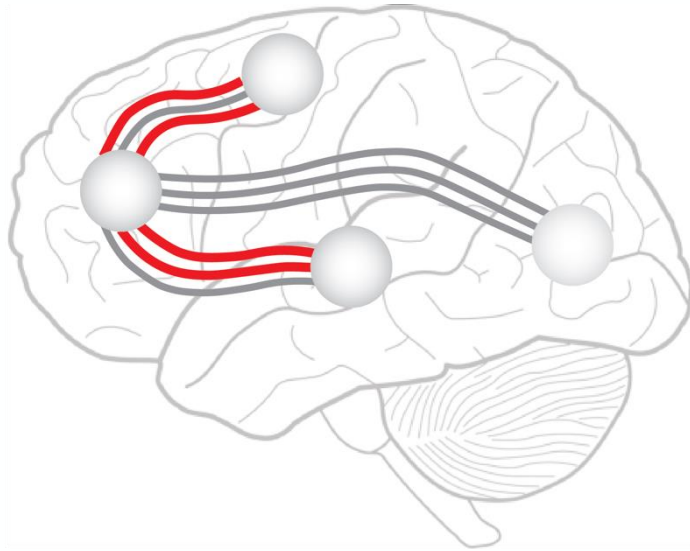


reading (visual)

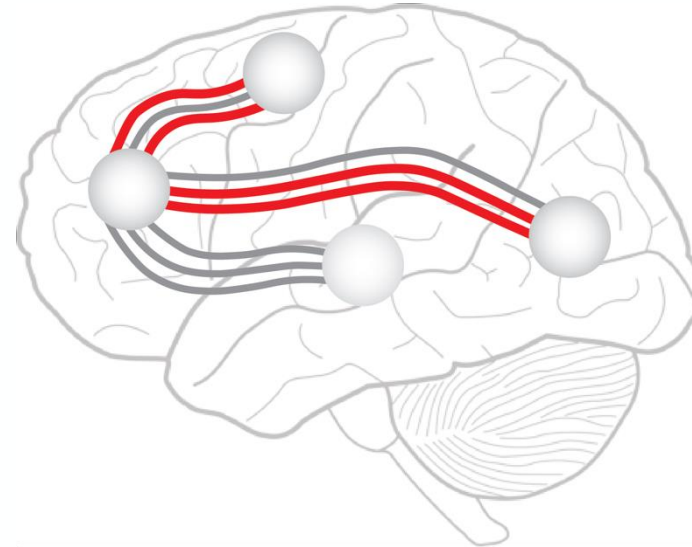


Dynamic Routing Behavior task

listening (auditory)



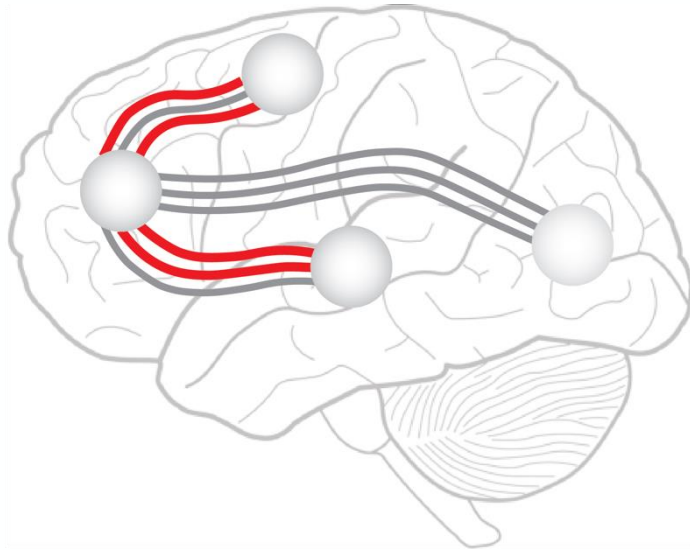
reading (visual)



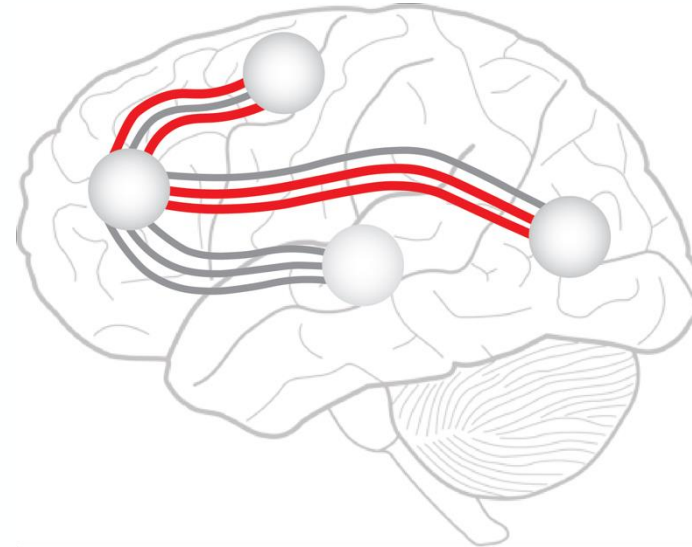
sensory switching task: train animal to switch between paying attention to an auditory stimulus OR a visual stimulus

Dynamic Routing Behavior task

listening (auditory)



reading (visual)

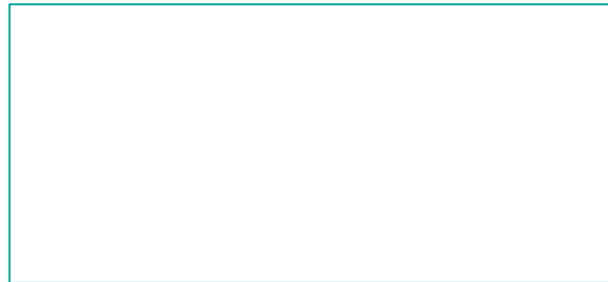
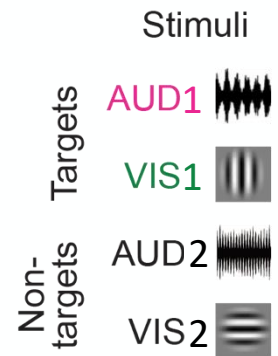


sensory switching task: train animal to switch between paying attention to an auditory stimulus OR a visual stimulus

How does **attention** modulate processing of sensory information?

Dynamic Routing Task: Stimuli

4 different stimuli: 2 visual, 2 auditory
VIS1, VIS2, AUD1, AUD2

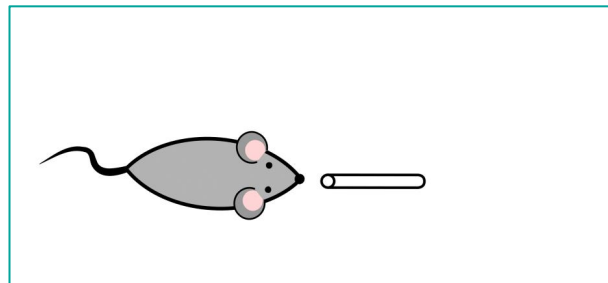


Dynamic Routing Task: Rewards

VIS1 and AUD1 associated with reward

VIS2 and AUD2 are never rewarded

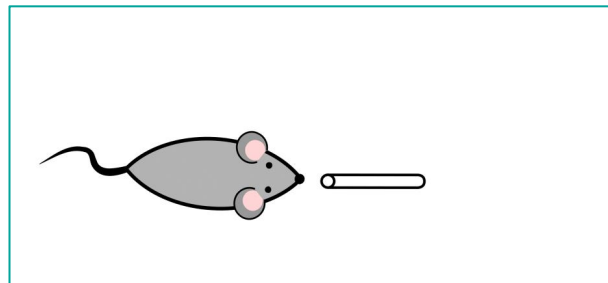
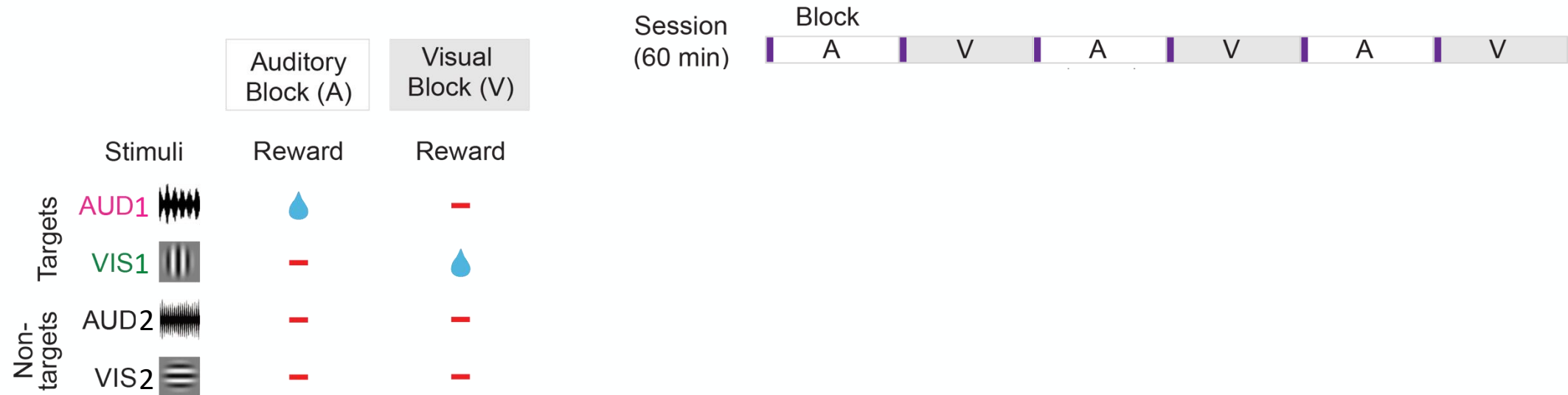
	Stimuli	Reward	Reward
Targets	AUD1 		
	VIS1 		
Non-targets	AUD2 		
	VIS2 		



Dynamic Routing Task: Context Switching

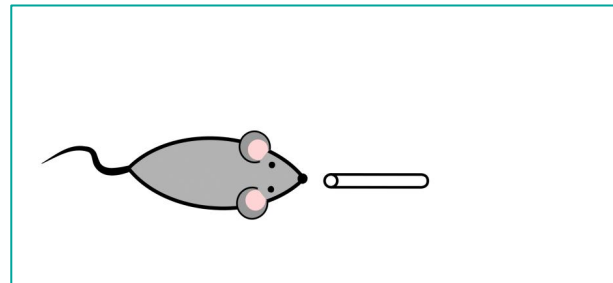
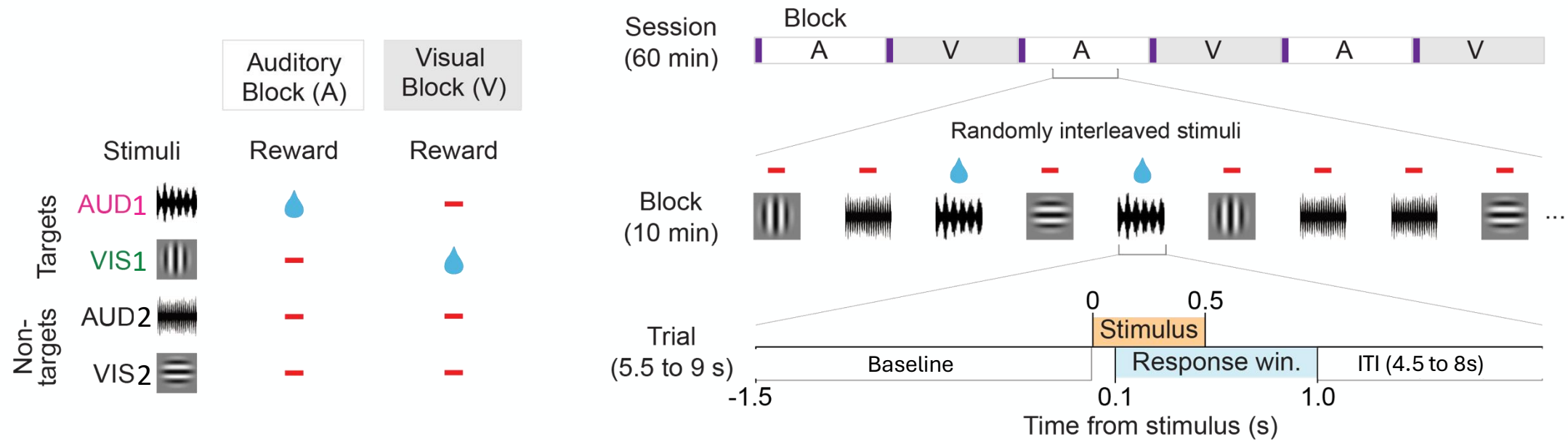
Rule changes:

trials organized into blocks when either auditory OR visual stimulus is rewarded



Dynamic Routing Task: Trial Structure

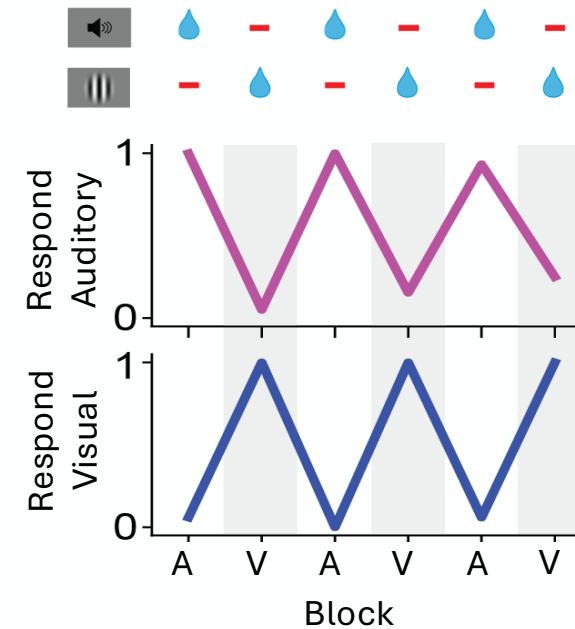
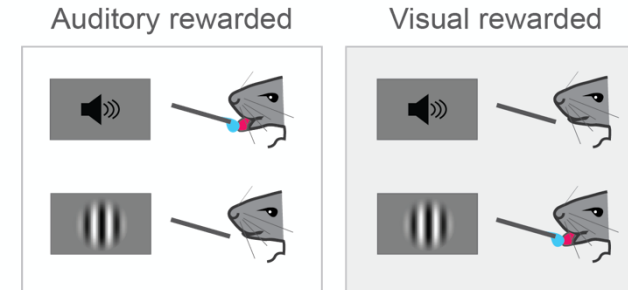
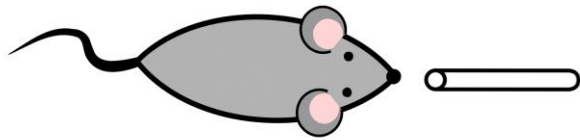
All 4 stimuli are presented in each block at random



Dynamic Routing Task

Block switches:

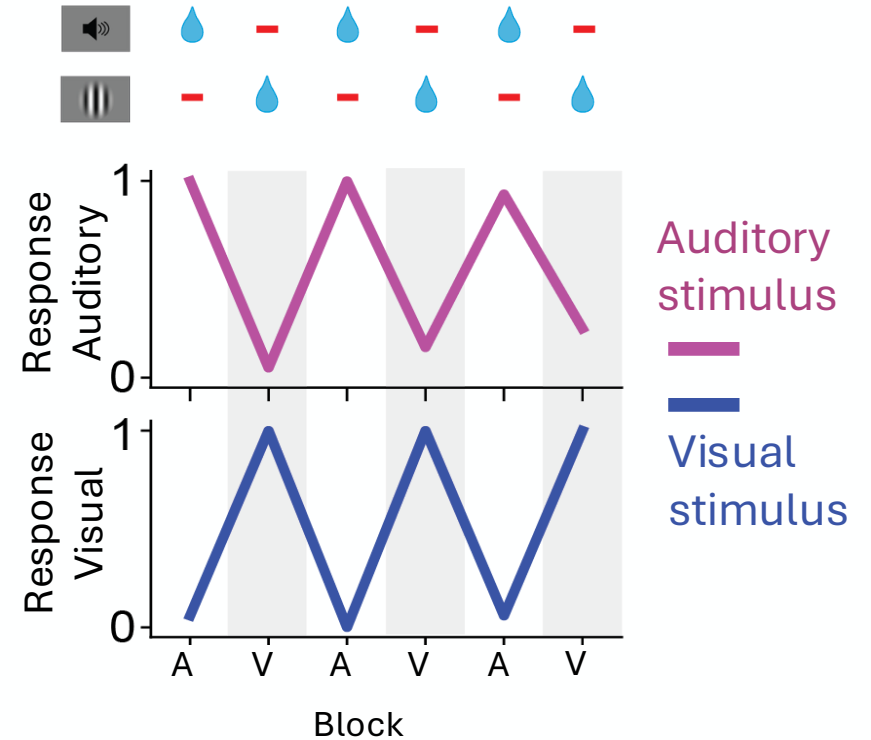
Animal trained to switch between listening vs looking for rewarded auditory and visual stimuli



many processes required for context-dependent behavior

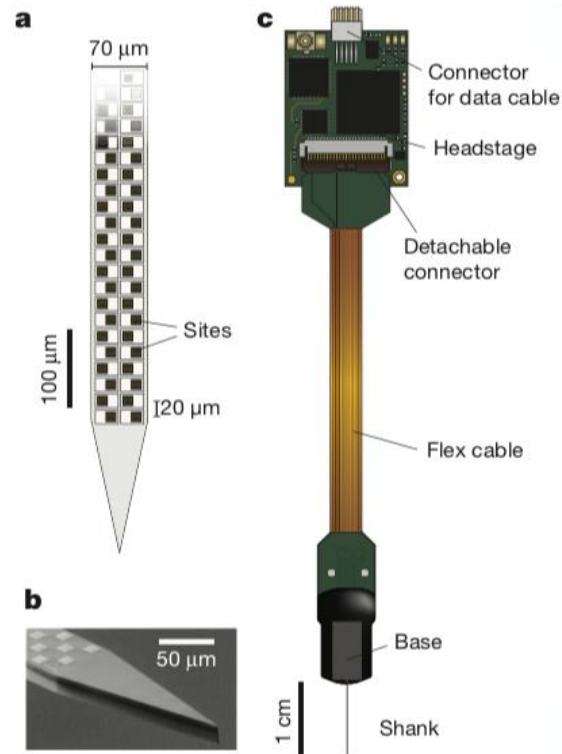
- sensory coding
- sensory -> motor transformations
- context / task rules
- maintenance / memory of context
- rule updating by cues and feedback
- context-dependent decision-making

→ necessitates a brain-wide approach



Methods for brain-wide neural activity recording: extracellular electrophysiology

Neuropixels probe



Neuropixels (NP) probes offer a high spatiotemporal resolution and large volume coverage

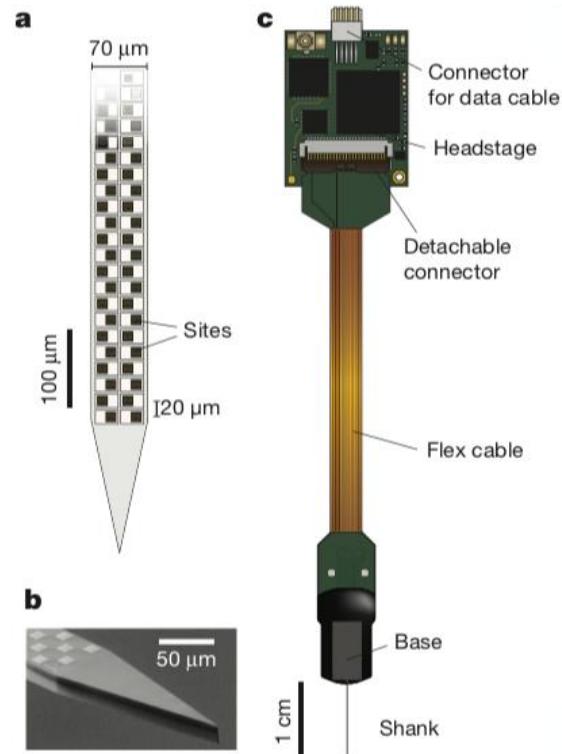
NP 1.0:

- 384 12 x 12 μm electrodes
- Checkerboard pattern
- Acquisition rate: 30 kHz

Jun et al. (2017) *Nature*

Methods for brain-wide neural activity recording: extracellular electrophysiology

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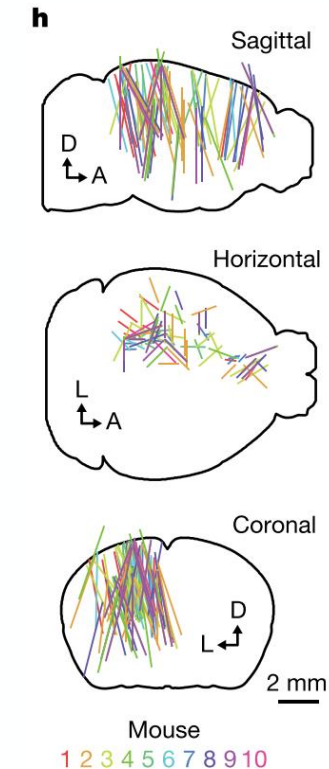


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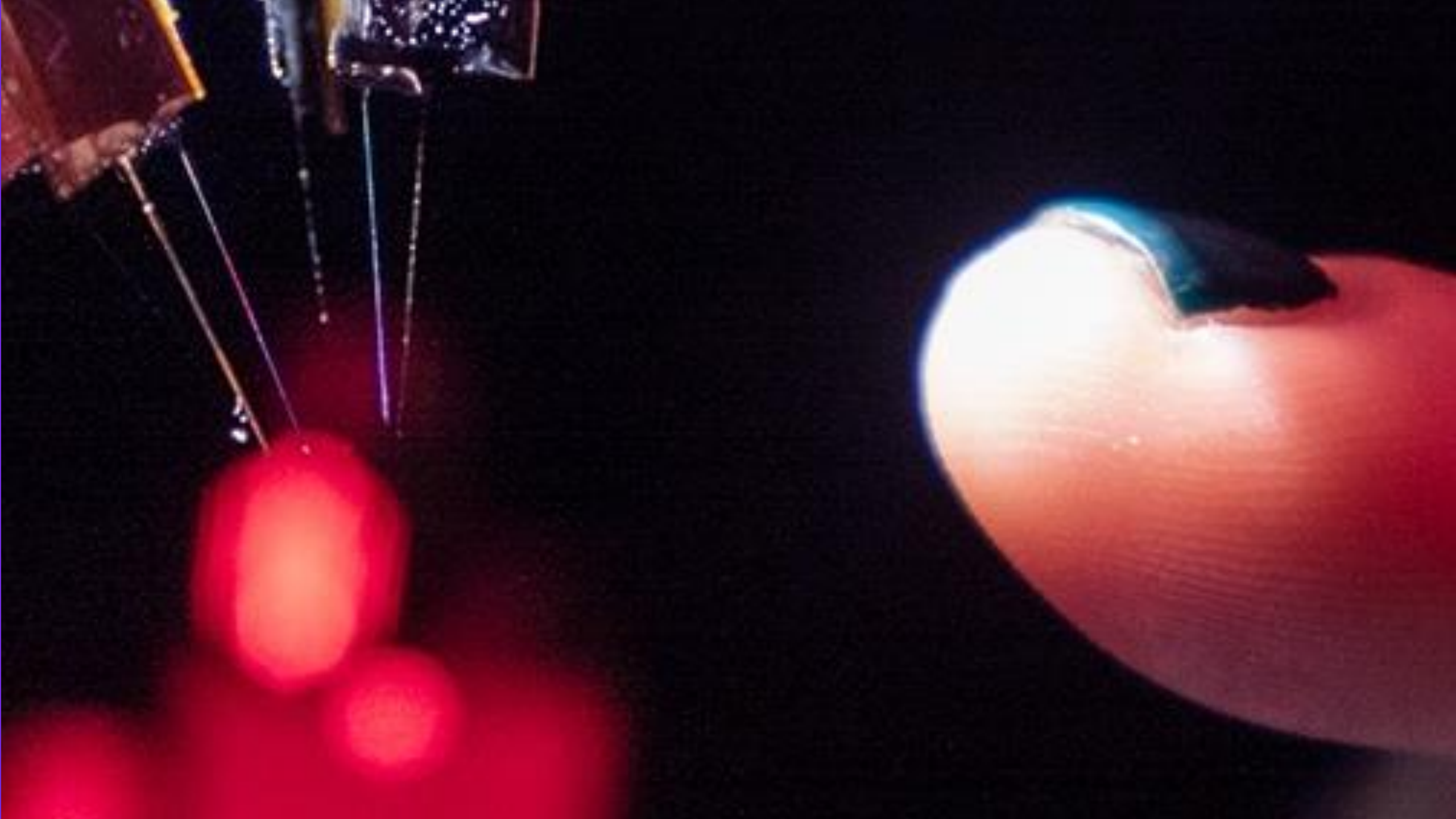
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NP 1.0:

- 384 12 x 12 μm electrodes
- Checkerboard pattern
- Acquisition rate: 30 kHz
- Record 100s–1000s of neurons across multiple brain regions simultaneously
- Resolve single spikes (action potential from a neuron)



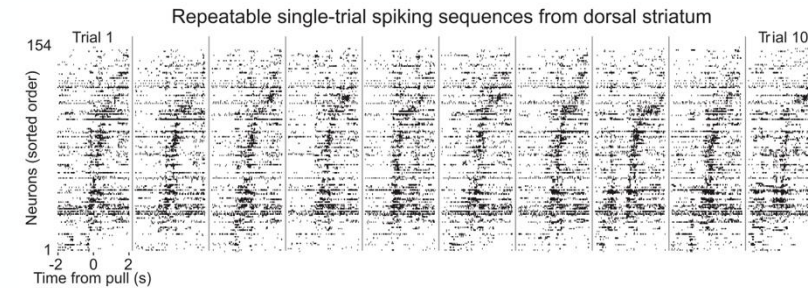
Steinmetz et al., (2019) *Nature*



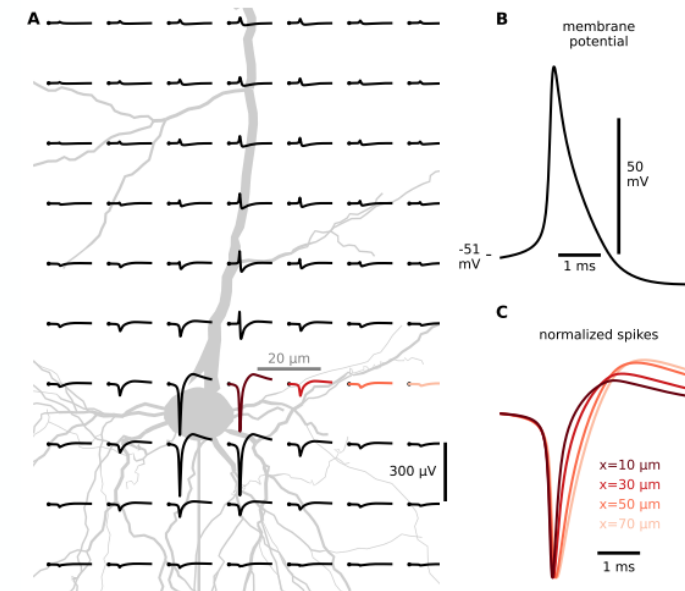
Experimental methods: extracellular electrophysiology

Spikes

- Compatible with awake, behaving animals, enabling direct links between neural dynamics and behavior
- Data scale allows testing of population-level theories: distributed coding, dimensionality reduction, neural manifolds, predictive processing
- A critical tool for bridging computational models and biological reality



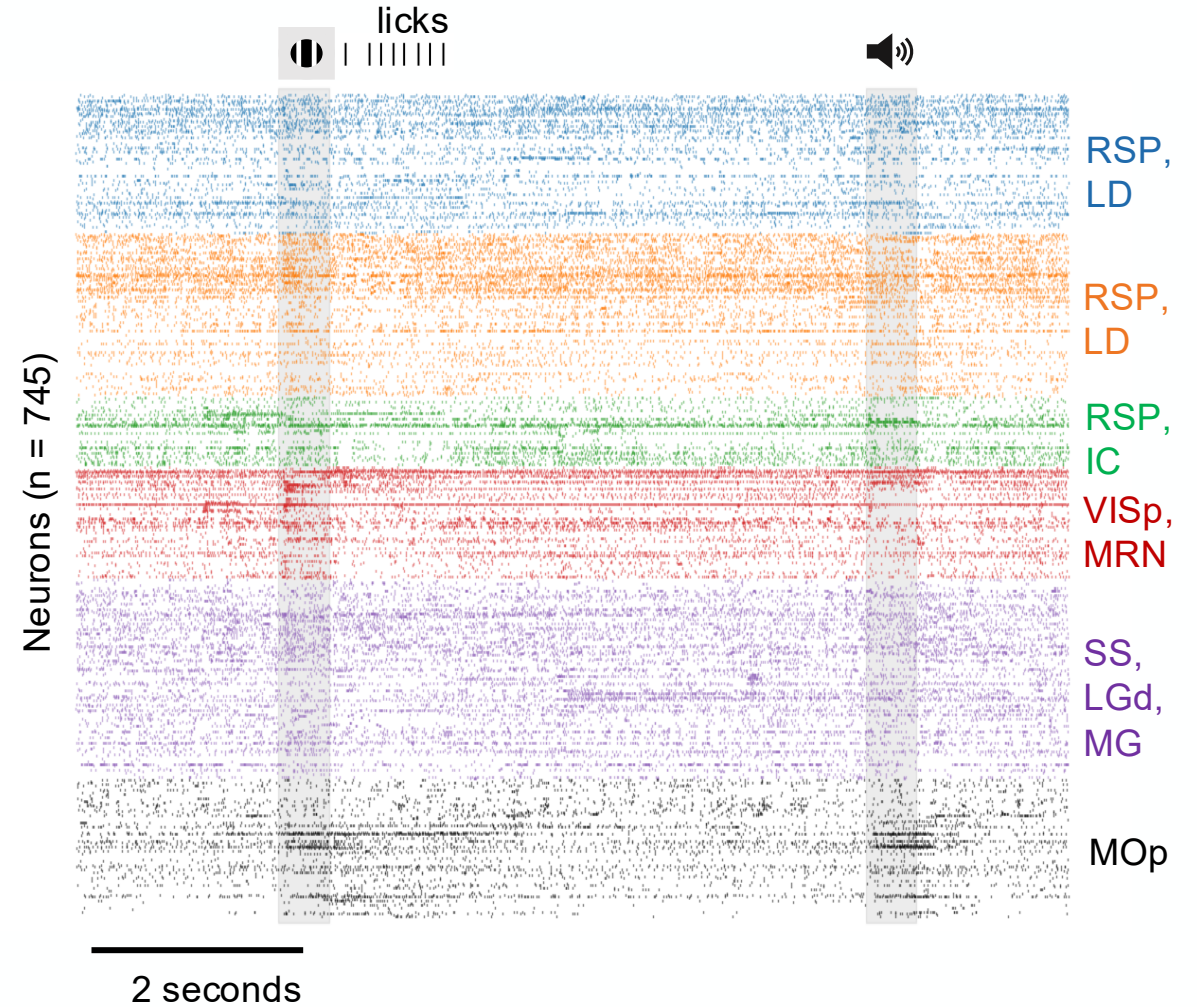
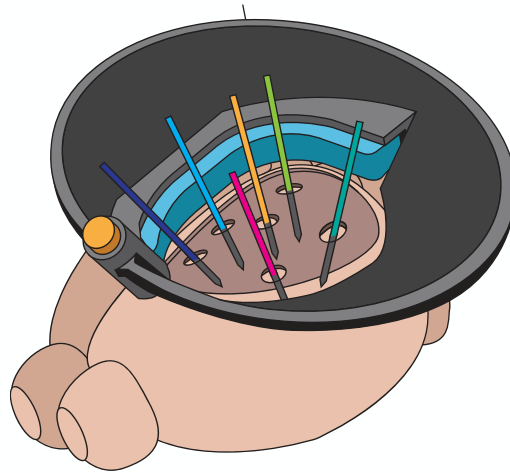
Steinmetz et al., (2021) *Science*



Electric Brain Signals, (2024)

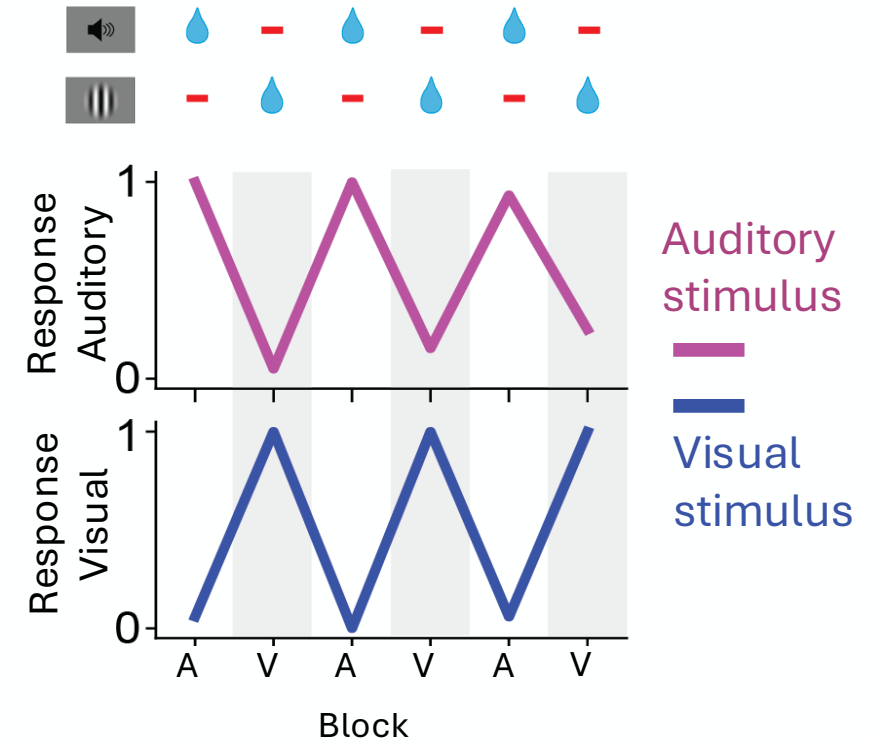
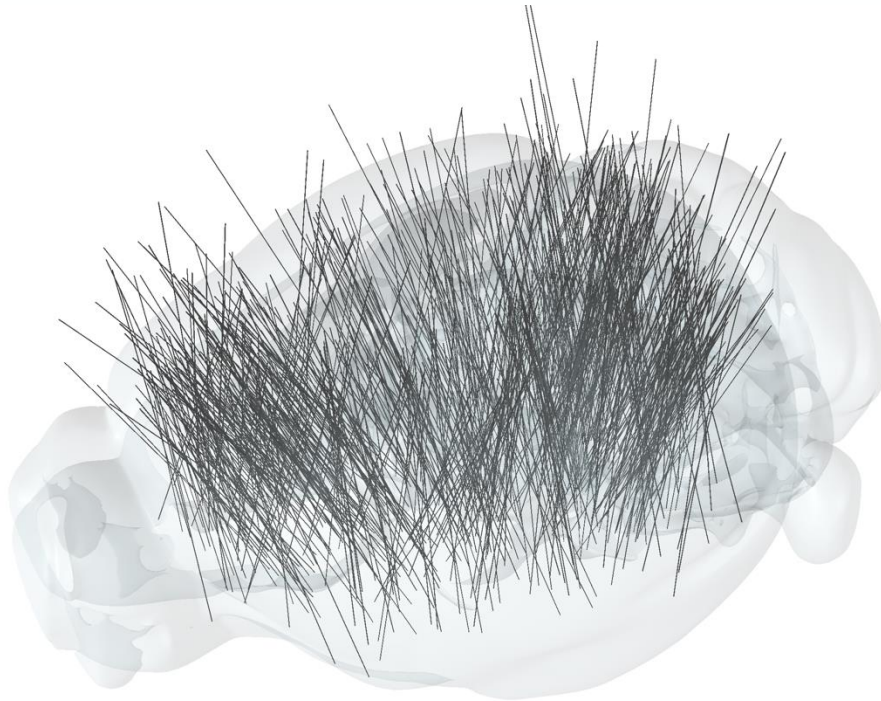
multi-regional neuropixels recordings

6 probes recording simultaneously



Spiking activity organized by brain region
Each row = 1 unit (neuron)

Dynamic Routing Dataset



>100k neurons, 836 insertions, 170 sessions, 53 mice, ~150 areas sampled

What brain regions were recorded from?





What brain regions were recorded from?

In the dataset you'll be working with, ~40 brain regions consistently sampled with good quality data

Isocortex

Sensory cortex: processes sensory information

Visual

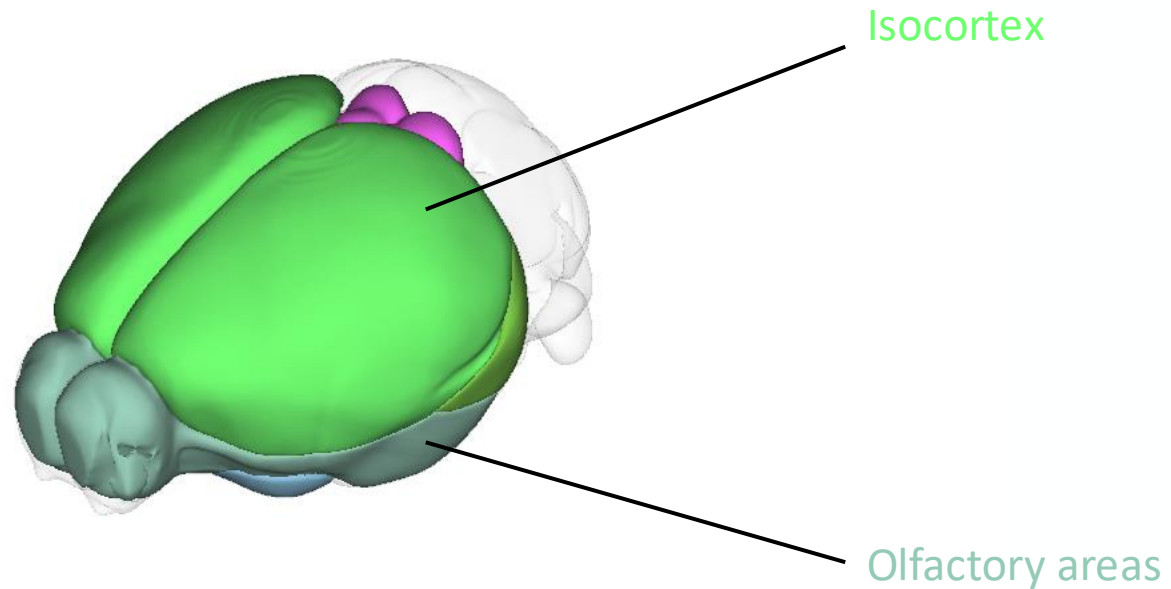
Auditory

Olfactory

Motor + somatosensory cortex

Prefrontal cortex:

- complex cognitive tasks
- decision making



Isocortex

Sensory cortex: processes sensory information

Visual

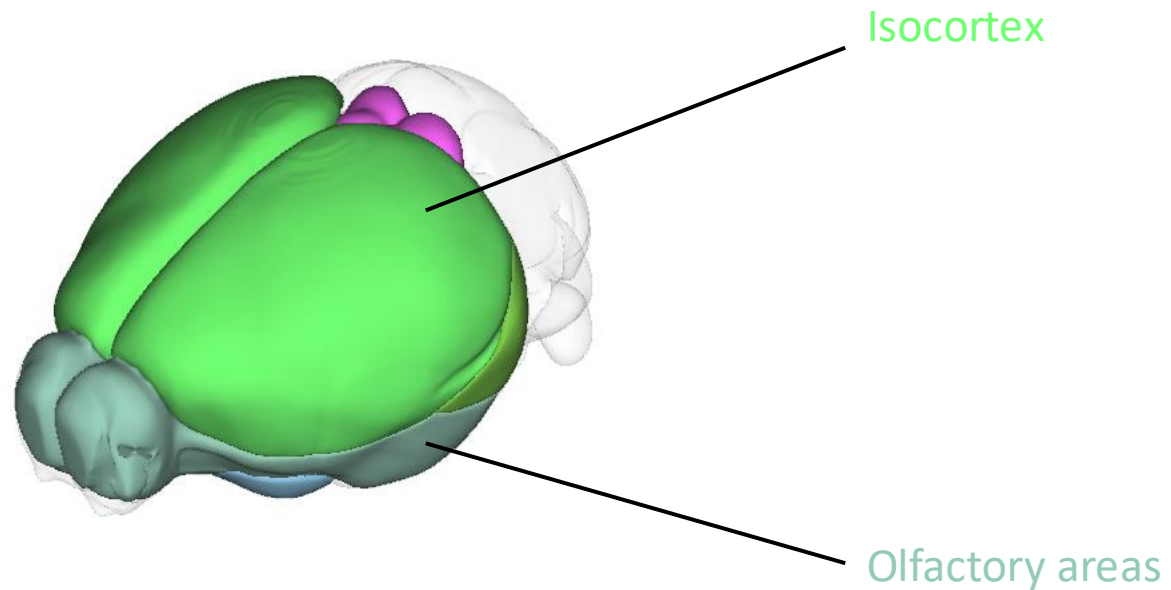
Auditory

Olfactory

Motor + somatosensory cortex

Prefrontal cortex:

- complex cognitive tasks
- decision making

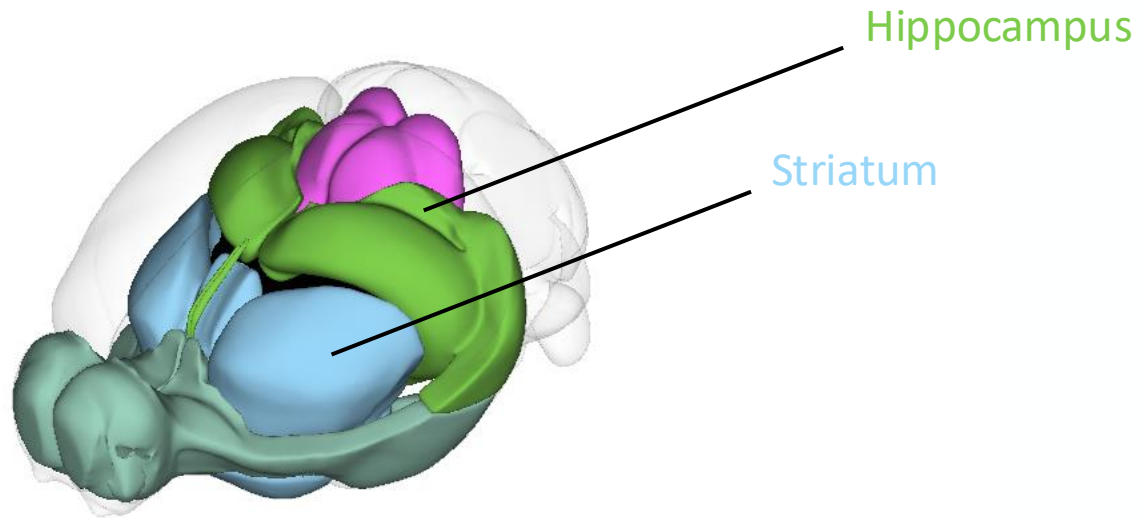


3 mm

Subcortical and deep brain structures

Hippocampus:
memory and learning

Striatum:
motor control, habits,
cognitive behaviors



3 mm

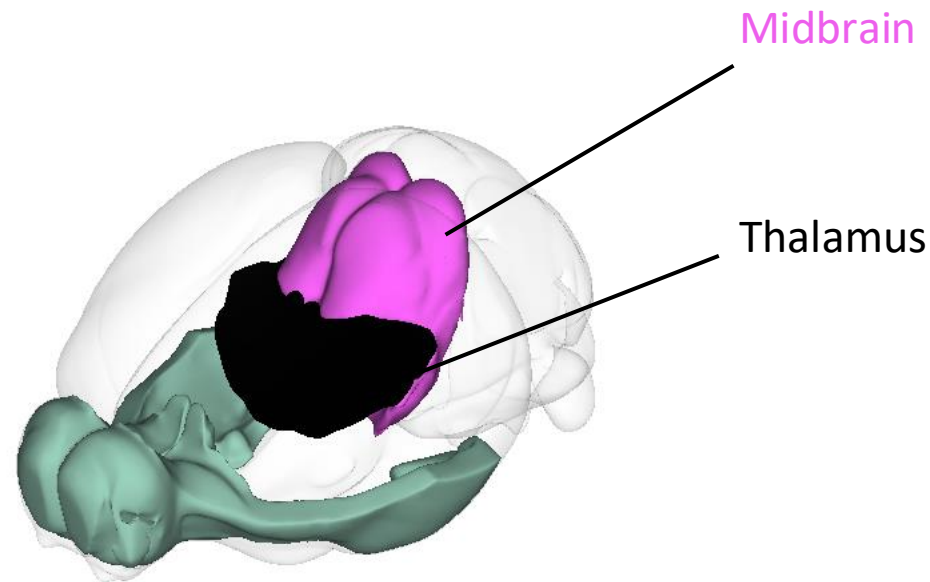
Subcortical and deep brain structures

Midbrain:

- part of brainstem
- relay muscle movements
- neuromodulation centers

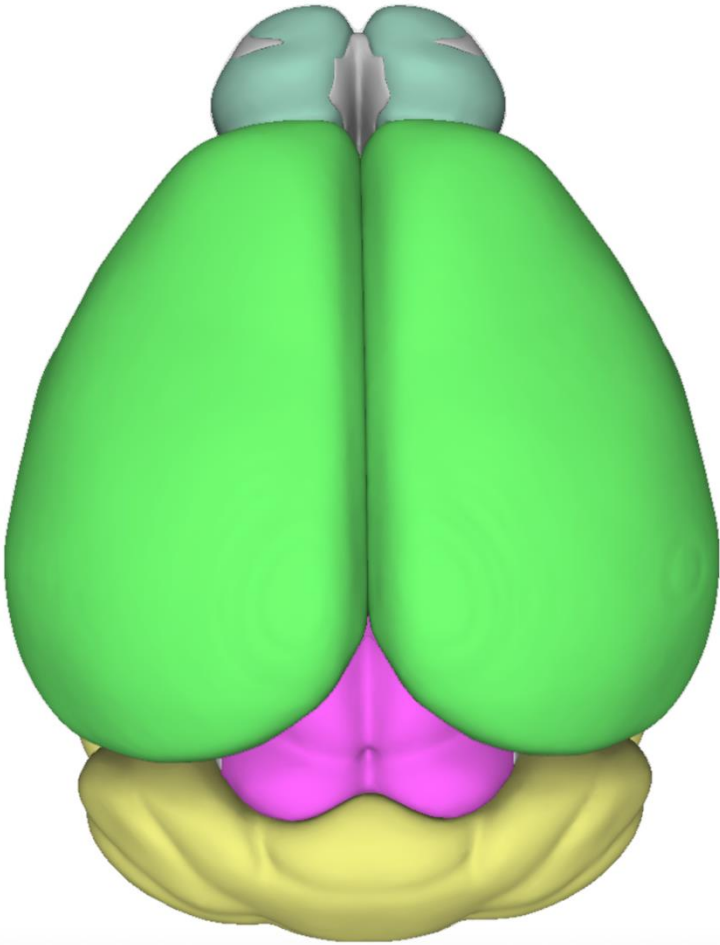
Thalamus:

- major input/output to cortex
- sensory, motor, and higher order thalamic nuclei

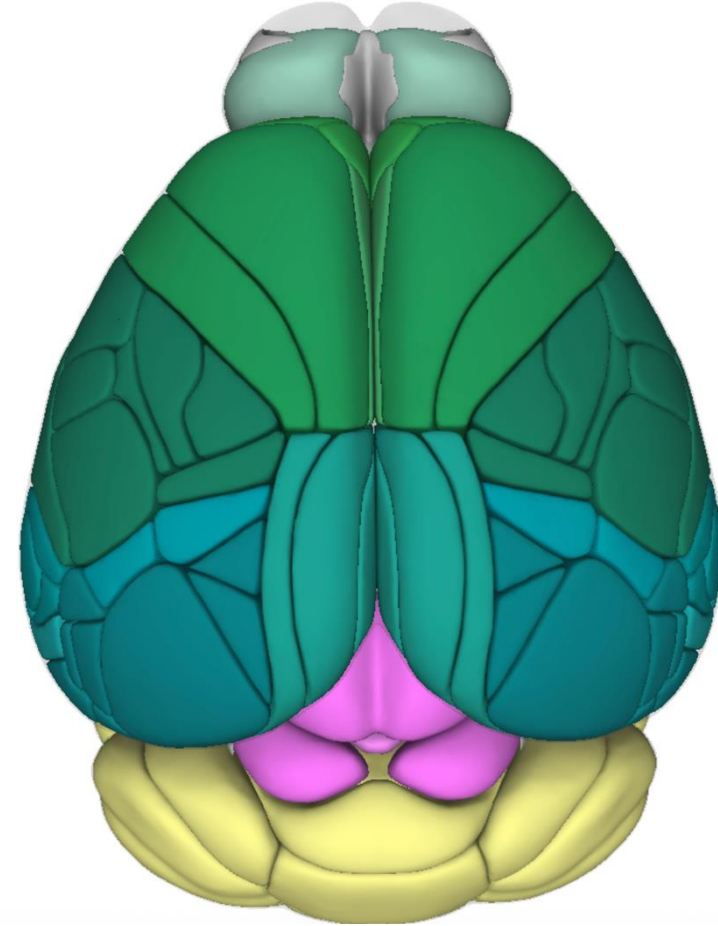


3 mm

Mouse brain–Allen Common Coordinate Framework



11 divisions



351 structures
611 sub-structures

Resource for exploring CCF brain structures : <https://spelunker.cave-explorer.org/#!middleauth+https://global.daf-apis.com/nglstate/api/v1/6082312788770816>

What brain regions were recorded from?



Cerebral Cortex (Isocortex)

Six-layer outer sheet — sensory, motor, and cognitive areas.

FRP – Frontal pole • ACAd/ACAv – Anterior cingulate, dorsal/ventral • MOs – Secondary motor • MOp – Primary motor • SSp – Primary somatosensory • SSs – Supplemental somatosensory • ORBv/ORBl/ORBm – Orbital areas • PL – Prelimbic • ILA – Infralimbic • VISp – Primary visual (+ VISal/VISam/VISrl/VISli/VISpm) • AUDp – Primary auditory (+ AUDd/AUDv/AUDpo) • TEa – Temporal association • ECT – Ectorhinal • PERl – Perirhinal • RSPd/RSPv – Retrosplenial • GU – Gustatory



Hippocampal Formation (HPF)

Spatial mapping and episodic memory, folded beneath the cortex.

DG – Dentate gyrus • CA3 – Cornu Ammonis 3 • CA2 – Cornu Ammonis 2 • CA1 – Cornu Ammonis 1



Olfactory Areas (OLF)

Smell processing

MOB – Main olfactory bulb • AON – Anterior olfactory nucleus • TTd – Taenia tecta, dorsal • DP – Dorsal peduncular area



Thalamus (TH)

Central relay.

AD – Anterodorsal nucleus • AV – Anteroventral nucleus



Striatum & Septum (STR)

Action selection, habits, and motivation (dorsal + ventral, plus septum).

CP – Caudoputamen • LSR/LSc – Lateral septal nucleus, rostral/caudal



Midbrain (MB)

Orienting movements, neuromodulation, reflexive sensorimotor behavior.

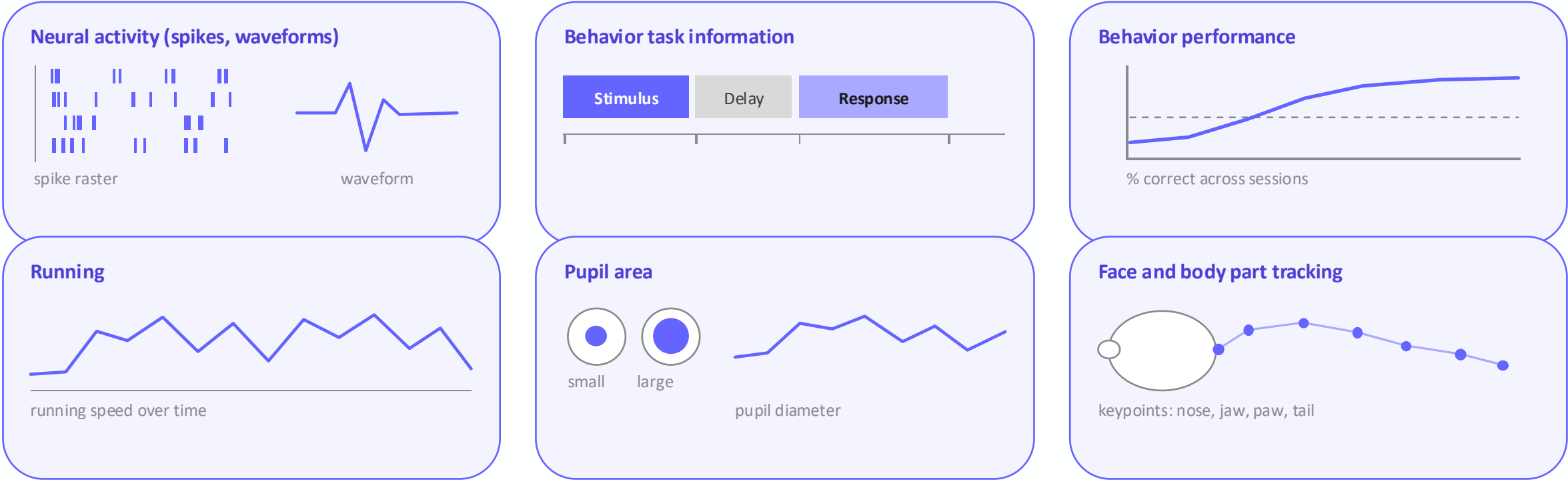
SCzo/SCsg/SCop – Superior colliculus, sensory layers (zonal / superficial gray / optic) • SCig/SCiw/SCdg/SCdw – Superior colliculus, motor layers (intermediate/deep gray & white) • PAG – Periaqueductal gray • PPT – Posterior pretectal nucleus

What data is available?

Metadata

Subject	Session	Rig	Genotype
mouse 664851	2024-07-12	Ephys rig 3	Sst-IRES-Cre
mouse 671093	2024-07-19	Ephys rig 1	Vip-IRES-Cre

Neurodata Without Borders files



Data you'll learn to work with

Metadata

Subject	Session	Rig	Genotype
mouse 664851	2024-07-12	Ephys rig 3	Sst-IRES-Cre
mouse 671093	2024-07-19	Ephys rig 1	Vip-IRES-Cre

Neurodata Without Borders files

Neural activity (spikes, waveforms)

spike raster waveform

Behavior task information

Stimulus Delay Response

Behavior performance

% correct across sessions

Running

running speed over time

Pupil area

small large pupil diameter

Face and body part tracking

keypoints: nose, jaw, paw, tail



What scientific questions can we answer with this data?

Sensory responses across the brain:

- How is sensory information represented in different brain areas?

- Which areas have visual responses, auditory responses, mixed?

Cell types:

- What features of the task are neurons responding to?

- How are different cell types organized? For example, are there spatial clusters or gradients with respect to functional cell type profiles?

Context-dependent modulation

- How do different contexts and state affect neural activity? Some examples:
attention, reward history, errors, arousal



Corbett Bennett Sam Gale Tamina Ramirez Ben Oullette Arjun Sridhar Shailaja Akella Ben Hardcastle Ulises Pereira-Obilinovic



Ethan McBride Jackie Kuyat Alex Cahoon Pete Groblewski Hannah Cabasco Vayle LaFehr Mikayla Carlson

+ more



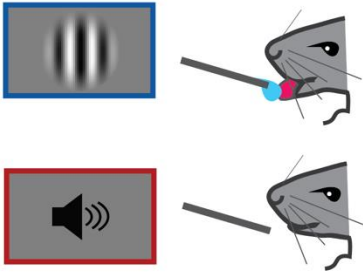
Shawn Olsen

R&D

Data collection,
analysis / modeling



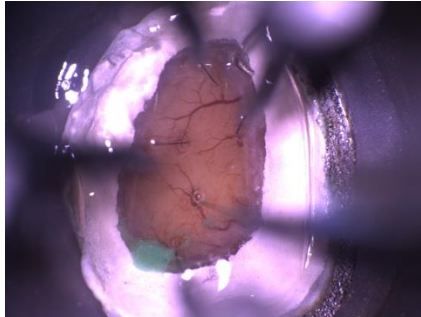
Behavior



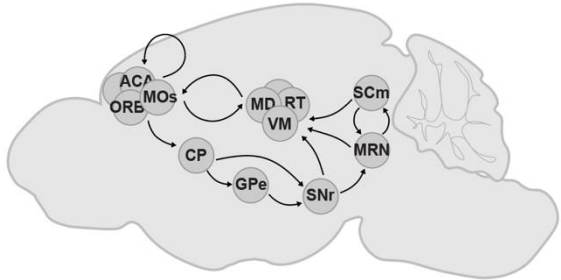
Surgical implant



Brain-wide recordings



Context network



Neural Dynamics Data Portal

data.allenneuraldynamics-test.org

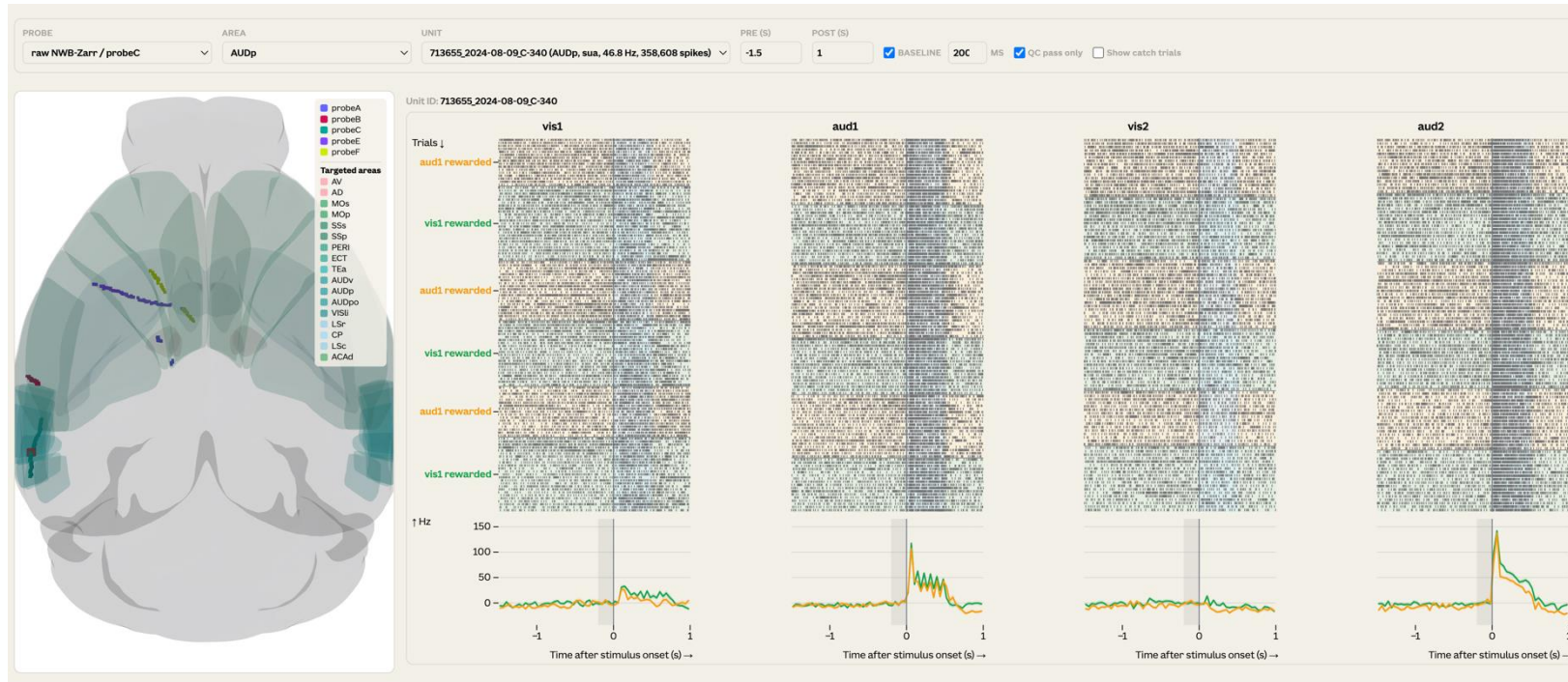
How to navigate to Dynamic Routing page

1. Upper right corner: Dashboard > SWDB
2. Click dynamic routing box
3. Select any session name



Neural Dynamics Data Portal

data.allenneuraldynamics-test.org



Task: Look at the raster plots and PSTHs for other units across probes

Can you find neurons with different response profiles? How would you describe them?

Some examples (visually responsive, stimulus suppressed, transient responses, context-dependent)



aleninstitute/

thank you/

Today's Code Tutorial

<https://github.com/leesuyee/dynamic-routing-tutorials>

Learning goals:

- How to work with large-scale, open neurophysiology data
- Create stimulus-aligned neural activity plots
- **Define cell types based on functional activity**
- **Quantify the effects of attention on sensory processing**



Resources

Github repo: (<https://github.com/leesuyee/dynamic-routing-tutorials>)

- Today's code tutorial

- Lecture slides

- README (overview of dynamic routing project, further references)

- Project ideas

- Starter code for project ideas (coming later this week)

Data Portal: data.allenneuraldynamics-test.org



Credits

Allen Institute intro slides courtesy of Lauren Alfiler

Allen Institute / Neural Dynamics slides courtesy of Saskia de Vries

Brain map, connectivity atlas, and cell types images and references (slide 12) courtesy of Jeremy Miller

Dynamic Routing slides courtesy of Shawn Olsen

Neuropixels background slides courtesy of Andrew Shelton